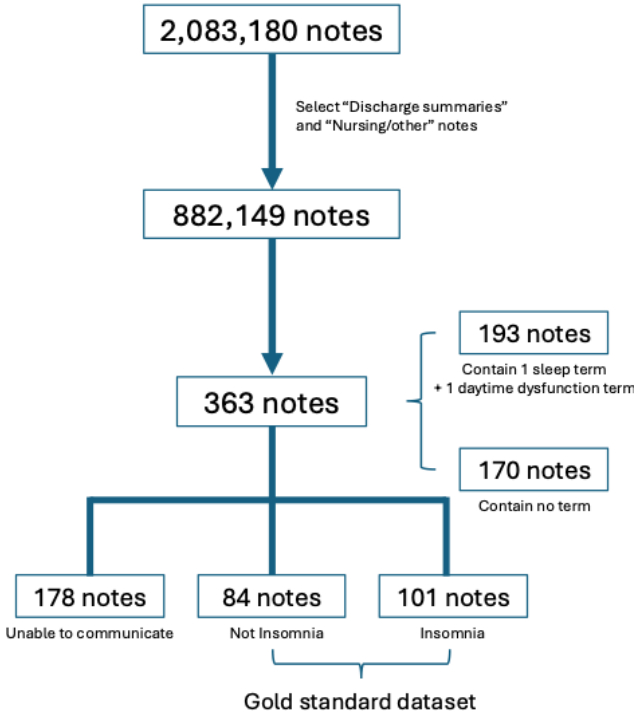

Figure S1: Overview of the clinical notes collection utilized for the development of our EHR-based insomnia phenotype.

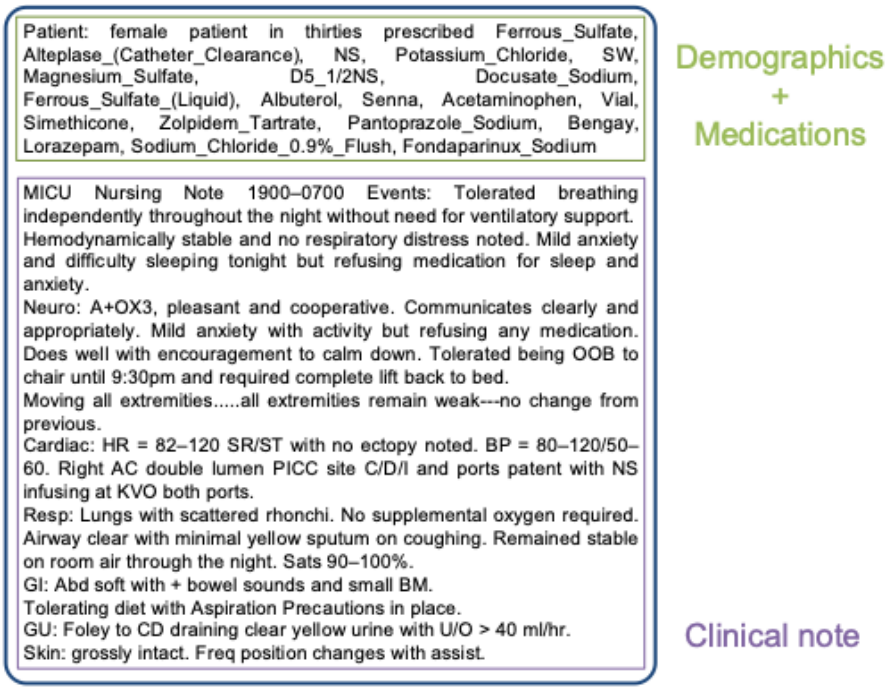

Figure S2: Each clinical note from MIMIC was enriched with both demographic and medication information.

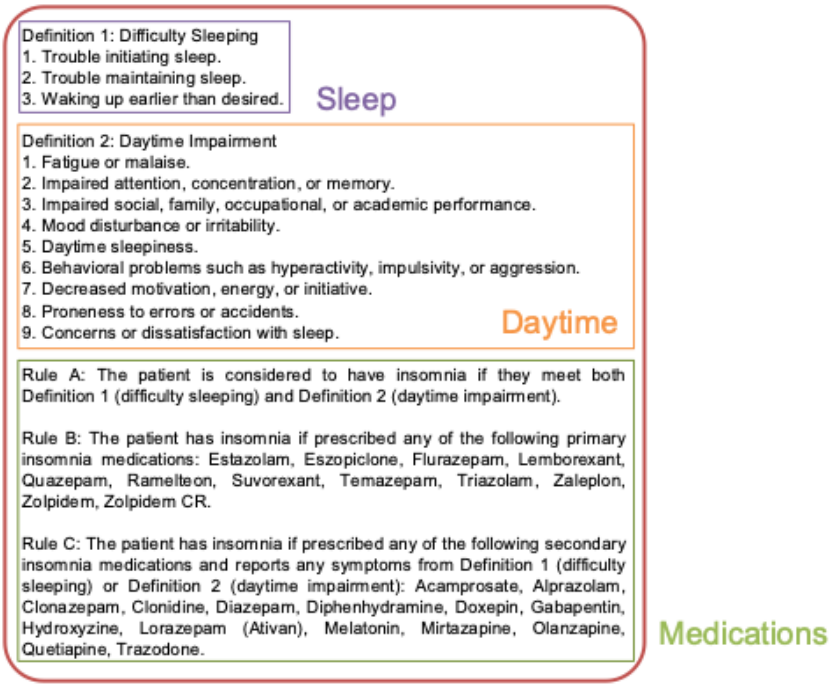

Figure S3: Illustration of the rules used to detect patients likely to suffer from insomnia.

ERROR ANALYSIS

To evaluate the limitations of the generative models, we conducted a qualitative error analysis of the top-performing models for each corpus (Llama 70B for MIMIC; Llama 405B for KUMC). Analysis focused on the most challenging rule components: Definition 2 and Rule C in the MIMIC corpus, and Definitions 1, 2, and Rule C in the KUMC corpus (see Table S1 below). Although Rule A was numerically the most challenging component, it was excluded from this specific analysis as its errors were entirely secondary to misclassifications in Definitions 1 and 2.

A recurring challenge across both datasets was the accurate identification of daytime impairment (Definition 2), which yielded among the lowest F1 scores. Qualitative analysis revealed that LLMs frequently misinterpreted non-specific symptoms—such as anxiety, fatigue, or cognitive deficits—by failing to account for the broader clinical context. For example, in the MIMIC corpus, cognitive symptoms following traumatic brain injury were incorrectly attributed to insomnia rather than the underlying neurological condition. Similarly, in the KUMC corpus, fatigue was often misidentified as an insomnia symptom despite plausible links to acute viral illness.

Medication misclassification also contributed to errors in Rule C. In the MIMIC corpus, models occasionally labeled medications used for acute ICU agitation (e.g., haloperidol,

midazolam) as secondary insomnia treatments. In the KUMC corpus, medications like tramadol or modafinil were sometimes erroneously categorized as insomnia treatments. Finally, in outpatient notes, the models occasionally failed to recognize implicit references to sleep difficulty, such as treating "sleep disturbance" as evidence of daytime impairment but not as difficulty initiating or maintaining sleep.

Corpus	Rule component	FP	FN	Clinical note	LLM response	Error category
MIMIC	Definition 2	16	6	*a/p: 27 yo male s/p bilat SAH, subfalcine and uncal herniations w/ large initial midline shift*	The patient reports symptoms related to daytime impairment: "short term memory deficit suspected", and "flat affect"	Daytime impairment related to external cause (brain herniation)
	Rule C	6	3	*male patient in fifties prescribed Sucralfate, Glycopyrrolate, ... Midazolam HCL, ... Haloperidol*	The patient is prescribed Midazolam and Haloperidol and has difficulty sleeping (Definition 1) and daytime impairment (Definition 2); therefore, Rule C is satisfied	Incorrectly identifying secondary insomnia medications
KUMC	Definition 1	4	6	*ROS: All 14 systems reviewed and found to be negative except as above and as follows. + activity change, sweating, fatigue, sleep disturbance*	The patient reports symptoms related to daytime impairment: "fatigue" and "sleep disturbance"	Missing sleep difficulty symptoms
	Definition 2	22	8	*Diagnoses and all orders for this visit: Viral Illness Cough ... Constitutional: Positive for fatigue and fever*	The patient reports symptoms related to daytime impairment: "fatigue"	Daytime impairment related to external cause (viral illness)
	Rule C	12	9	*- zolpidem (AMBIEN) 5 mg tablet Take one tablet by mouth daily*	The patient is prescribed Zolpidem and has difficulty sleeping (Definition 1) and daytime impairment (Definition 2); therefore, Rule C is satisfied	Incorrectly identifying primary insomnia medication as secondary

Table S1: Representative errors made by the best-performing generative model for each corpus (Llama 70B for MIMIC, Llama 405B for KUMC) across the most challenging rule components. FP = false positives, FN = false negatives.

DATA COLLECTION

<u>Sleep disturbance terms</u>

adjustment insomnia
awakening early
awakening early morning
bad dream
bad dreams
bizarre dreams
bothered by difficulty sleeping

broken sleep
cannot get off to sleep

cannot sleep at all
chronic insomnia

complaining of insomnia

complaining of nightmares
complaining of vivid dreams
delayed onset of sleep

difficulty staying or falling asleep

difficult sleeping

difficulty sleeping through the night

difficulties sleeping
difficulty falling asleep
difficulty getting to sleep
difficulty in sleep initiation
Difficulty in sleep maintenance
difficulty in sleep maintenance

difficulty sleep
difficulties sleep
difficulty sleeping
difficulties sleeping
difficulty staying asleep
disturbance in sleep
disturbance of sleep behavior

disturbance in sleep
disturbed sleep
disturbances sleep
disturbances of sleep
dysnystaxis
early awakening
early morning awakening
early morning waking
early waking
fitful sleep
frequent night waking
frightening dreams
have nightmares
hyposomnia
impaired sleep
inability to sleep
initial insomnia
insomnia
insomnia matutinal
insomnia chronic
insomnia late

Insomnia unspecified
insomnia vesperal
interrupted sleep
interrupting sleep
keeps waking up
late insomnia
light sleep
light sleeping
lights sleeping
hard to sleep through night
hard to sleep through the
matutinal insomnia
middle insomnia
my sleep was restless
nightmares
nightmares
night wake
night wakes
night waking
not getting enough sleep
organic insomnia
other insomnia
persistent insomnia
poor quality sleep
poor sleep
poor sleep pattern
primary insomnia
problems with sleeping
prolonged sleep
restless sleep
restless sleeping
short of sleep
short sleeping
sleep deficit
sleep deprivation
sleep difficult
sleep difficulties
sleep disorder insomnia
sleep disorder insomnia
sleep disorders
sleep disturbance
sleep disturbances
sleep disturbed
sleep dysfunction
sleep is restless
sleep was restless
sleeping difficulty
sleeping difficulties
sleeping disturbances
sleep restless

sleeplessness
terrifying dreams
terminal insomnia
tosses and turns in sleep
trouble falling asleep
trouble sleeping
unable to sleep
unpleasant dreams
vesperal insomnia
vivid dreams
wakes and cannot sleep again
wakes early
waking during night
waking too early
waking night
wakes up at night
wakes up during night

Daytime dysfunction terms

aggression
aggressive
Aggressive behavior
aggressive behaviors
aggressive behaviour
aggressive behaviours
Aggressive behavior
violent behavior
aggressiveness
behavior aggressive
violent aggressive behavior
anger
anger outburst
anger outbursts
Anger reaction
Angers
angry
angry mood
Difficulty controlling anger
difficulty managing anger
Feeling angry
Irritability and anger

Mad
mood angry
Outburst of anger
outbursts of anger
Outbursts of rage
nervous tension
angst
anxiety
Anxiety
anxiety chronic
anxiety disorder
Anxiety disorder
Anxiety NOS
anxiety state
Anxiety state NOS
Anxiety state unspecified
Anxiety state, unspecified
Anxiety states
anxious
Anxious behavior
anxious mood
anxiousness
Feel anxious
feeling anxious
Feeling anxious
feeling nervous
Feeling of Nervousness
Mild anxiety
mood anxious
Nervously anxious
nervousness
Nervousness
Unspecified anxiety disorder
anxiety depression mixed
anxiety with depression
Decreased Mood
depressed
depressed mood

depression anxiety
depression mild
depression minimal
depression symptom
depression with anxiety
depressions symptoms
depressive disorder
Depressive disorder NOS
depressive symptoms
emotional upset
feeling down
Feeling low
Feeling of sadness
Low mood
Mild depression
Minimal depression
miserable
Miserable
Mixed anxiety and depressive condition
Mixed anxiety and depressive disease
Mixed anxiety and depressive disorder
mixed anxiety depression
Morose mood
Sad
Sadness
dysphoria
dysphoric mood
mood dysphoric
Excessive somnolence
somnolence excessive
malaise and fatigue
Complaining of "tired all the time"
Decrease in energy
decreased energy
demonstrates lethargy
easily tired
easily tires
easily tiring

Energy decreased
energy loss
exhaustion
Exhaustion, NOS
fatigue
Fatigue (lassitude)
fatigue malaise
Fatigue NOS
Fatigued
fatigues
fatiguing
feel tired
feeling tired
feeling tired all the time
feeling tired or poorly
general malaise
having little energy
lack of energy
Lack of vitality
lacking energy
Lacking in energy
lethargic
Lethargy
Malaise
Malaise and fatigue
Rapidly tired
Sometimes tired
tatt
TATT
TATT - Tired all the time
tired
Tired
tired all the time
Tired all the time
tired feeling
Tired out
Tiredness
tires easily

Tires quickly
tiring easily
Torpid
Washed-out
Washed out
weariness
worn out

feel bad
Feeling bad
Feels bad

affect unhappy
Feeling sad
Feeling unhappy
unhappy

homicidal tendencies

Behavior hyperactive
behavior; overactivity
Behaviour hyperactive
hyperactive
hyperactive behavior
Hyperactive behaviour
hyperactivity
Increased purposeful goal-directed activity
More active than typical
motor hyperactivity
Motor Hyperactivity
overactivity
Impulsive
impulsiveness
feeling irritable
Feeling irritated
feeling tense
Feeling tense
fussiness

irritability
irritable
Irritable
irritable mood
Irritable Moods
mood irritable
Mood, Irritable
tense
tense feeling
tenseness

decreased interest
lack of interest
loss of interest
absence of motivation
Decrease in motivation
Decreased motivation
Diminished motivation
lack of motivation
lacking motivation
loss motivation
Loss of motivation
Low motivation
Motivational deficit
unmotivated
bad memory
difficulty remembering
disturbed memory
disturbed concentration
memory disturbance
Disturbance of memory
dysmnesia
Dysmnesia
forgetful
forgetfulness
impair memory
impaired memory
impairment memory

LOM - Loss of memory
loss memory short-term
loss memory short term
loss of memory
memory deficit
memory deficits
memory disturbance
Memory dysfunction
Memory impaired
memory impairment
Memory impairment
memory loss
memory loss or impairment
Memory Losses
memory poor
memory problem
memory problems
Memory problems
poor memory
Poor memory
Poor short-term memory
short-term memory loss
Short-term memory poor
short term memory loss
Altered mood
altered moods
Change in mood
Changeable mood
disorder mood
Disorder of affect
emotional instability
emotional lability
emotionally labile
Instability emotional
labile mood
Labile Moods
Lability emotional
mood change

Mood changes
mood disorder
mood disorder nos
Mood disorder, NOS
mood labile
mood lability
mood swing
mood swings
Mood variable
moody
Moody
Unspecified mood affective disorder
Unstable mood
Variable mood

fatigue
tiredness
tire easily
muscle weakness
weakness
attitude inattentive
concentrate inability
Concentration abnormal
concentration difficulties
concentration difficulty
Concentration impaired
Concentration impairment
concentration lack
concentration loss
difficulty concentration
Diminished ability to concentrate
disturbed mental concentration
General inattentiveness
impaired concentration
inability to concentrate
inability to maintain attention
inattention
inattentive

lack concentration
lack of concentration
Mental concentration difficult
Mental concentration difficulty
Mental concentration impaired
poor concentration
Poor concentration
trouble concentrating
unable to concentrate
daytime sleepiness
Daytime Sleepiness
daytime somnolence
Daytime Somnolences
drowsiness
Drowsiness
drowsiness was observed
drowsy
excessive sleep
excessive sleepiness
excessive sleepiness during day
excessive sleepiness during the day
daytime somnolence
Mental status, drowsy
sleepiness
Sleepiness
sleepiness daytime
Sleepiness, Daytime
sleepy
somnolence
Somnolence, Daytime
somnolent
Somnolent
emotional stress
Increased stress
Unspecified acute reaction to stress
physical aggression
Physical assaultive behavior
Physical assaultive behaviour

Physically assaultive behavior
Physically assaultive behaviour
Asthenia
asthenic
debility
Debility
Enfeebled
feel weak
feeling weak
Feeling weak
feeling weakness
feelings of weakness
feels weak
General weakness
generalized weakness
lassitude
weak
weak feeling
weakness
Weakness - general
Weakness generalised
Weakness generalized

ANNOTATION GUIDELINES

Insomnia Definition

According to the International Classification of Sleep Disorders-3rd Edition (ICSD-III), insomnia disorder can be classified as:

1. **Chronic Insomnia Disorder**

Patient experiences sleep disturbances for the last three months affecting the nighttime sleep for at least three times a week and affects the patients' overall functioning.

2. **Short-term Insomnia Disorder**

Sleep disturbances experienced for less than three months and affects the patients' overall functioning.

3. **Other Insomnia Disorder**

Sleep disturbances that do not meet the criteria for chronic insomnia or short-term insomnia disorder are classified under this category.

Annotation Schema

There are several different categories of information that the annotator should consider when tagging a note with Value of "Insomnia". Table 1 provides a listing of categories that should be considered during annotation.

Using these categories (Table 1) as a guideline, the annotator should use their best judgment and domain knowledge when assessing whether a note indicates insomnia.

Table 1.: Categories of text that that would assist in diagnosing insomnia.

Category	Details	Include	Exclude
Direct Symptoms	Patient reported symptoms	• Difficulty falling asleep at night • Waking up during the night • Waking up too early • Not feeling well-rested after a night's sleep • Daytime tiredness or sleepiness • Irritability, depression or anxiety • Difficulty paying attention, focusing on tasks or remembering • Increased errors or accidents • Ongoing worries about sleep	All other symptom mentions
Indirect Symptoms	Symptoms that are a consequence of the	• Fatigue or malaise.	Other indirect

	sleep continuity disturbance	• Attention, concentration, or memory impairment. • Impaired social, family, occupational, or academic performance. • Mood disturbance or irritability. • Daytime sleepiness. • Behavioral problems such as hyperactivity, impulsivity, and aggression. • Reduced motivation, decreased energy, or lack of initiative. • Proneness for errors or accidents. • Concern about sleep or dissatisfaction with sleep	symptom mentions
Diagnosis	Insomnia disorder diagnosis documented by provider	See list of ICD 9/10 diagnoses	
Comorbidities	Comorbid conditions/disorders that may contribute to insomnia disorder	• Alcoholism • Depression • Anxiety • Hyperlipidemia	
Medication	Mentions of medications that are prescribed to treat insomnia	• Estazolam • Flurazepam • Quazepam • Temazepam • Triazolam • Zolpidem • Zolpidem CR • Zaleplon • Eszopiclone • Ramelteon • Suvorexant	

		• Lemborexant • Doxepin • Trazodone • Quetiapine • Diphenhydramine • Olanzapine • Mirtazapine • Melatonin • Acamprosate • Clonidine • Gabapentin • Hydroxyzine • Alprazolam • Clonazepam • Lorazepam • Diazepam	

ICD-9 Codes for Insomnia

Insomnia, unspecified
780.52

Transient disorder of initiating or maintaining sleep
307.41

Persistent disorder of initiating or maintaining sleep
307.42

Persistent disorder of initiating or maintaining sleep
307.42

Lack of adequate sleep
V69.4

Insomnia due to medical condition classified elsewhere
327.01

Transient disorder of initiating or maintaining sleep
307.41

Persistent disorder of initiating or maintaining sleep
307.42

Organic insomnia, unspecified
327.00

Insomnia due to mental disorder
327.02

Other organic insomnia
327.09

insomnia with sleep apnea, unspecified
780.51

Other sleep disturbances
780.59

Dysfunctions associated with sleep stages or arousal from sleep
780.56

Sleep disturbance, unspecified
780.50

Alcohol induced sleep disorders
291.82

Drug induced sleep disorders
292.85

Nonorganic sleep disorder, unspecified
307.40

ICD-10 Codes for Insomnia

Adjustment insomnia
F51.02

Insomnia due to a medical condition
G47.01

Insomnia due to other mental disorder
F51.05

Insomnia unspecified
G47.00

Other insomnia
G47.09

Other insomnia not due to a substance of known physiological condition
F51.09

Other Sleep Disorders not due to A Substance or Known Physiological Condition
F51.8

Sleep Disorder not due to A Substance or Known Physiological Condition, unspecified
F51.9

Paradoxical insomnia
F51.03

Primary insomnia
F51.01

Psychophysiologic insomnia
F51.04

Alcohol use, unspecified with Alcohol-Induced Sleep Disorder
F10.982

Cocaine Abuse with Cocaine-Induced Sleep Disorder
F14.182

Cocaine Dependence with Cocaine-Induced Sleep Disorder
F14.282

Cocaine use, unspecified with Cocaine-Induced Sleep Disorder
F14.982

Opioid abuse with opioid-induced sleep disorder
F11.182

Opioid dependence with opioid induced sleep disorder
F11.282

Opioid abuse, unspecified with opioid-induced sleep disorder
F11.982

Other Psychoactive Substance Abuse with Psychoactive Substance-Induced Sleep Disorder
F19.182

Other Psychoactive Substance Dependence with Psychoactive Substance-Induced Sleep Disorder
F19.282

Other Psychoactive Substance use, unspecified with Psychoactive Substance-Induced Sleep Disorder
F19.982

Sleep deprivation
Z72.820

Sedative, Hypnotic or Anxiolytic Dependence with Sedative, Hypnotic or Anxiolytic-Induced Sleep Disorder F13.282

Sedative, Hypnotic or Anxiolytic use, unspecified with Sedative, Hypnotic or Anxiolytic-Induced Sleep Disorder F13.982

Sedative, Hypnotic or Anxiolytic use, unspecified with Sedative, Hypnotic or Anxiolytic-Induced Sleep Disorder F13.982

Insufficient Sleep Syndrome
F51.12